

Notes: Derrien & Kecskes (JF, 2013)

The Real Effects of Financial Shocks: Evidence from Exogenous Changes in Analyst Coverage

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1 Big picture

- **Question.** Does a financial-information shock have real effects on firms' investment and financing policies?
- **Shock.** The paper studies firms that lose analyst coverage because their covering broker closes or merges. The identifying idea is that broker disappearances remove information production for reasons largely unrelated to the covered firm's own investment plans.
- **Mechanism.** Losing an analyst increases information asymmetry between firms and outside investors. Higher information asymmetry raises the cost of external capital, reducing both investment and external financing.
- **Empirical design.** The authors use a matched difference-in-differences design. Treatment firms lose an analyst; control firms are matched by industry, total assets, Tobin's Q , cash flow, and analyst coverage.
- **Sample.** The sample includes 52 broker disappearances from 1994 to 2008: 19 broker closures and 33 broker mergers. These events cause 1,724 treatment firms to lose an analyst. Each treatment firm is matched to one control firm.
- **Main result.** Relative to matched control firms, firms that lose an analyst reduce total investment by 1.92% of total assets and total financing by 2.02% of total assets.
- **Investment margins.** Capital expenditures fall by 0.67%, research and development by 0.21%, and acquisitions by 0.97% of total assets.
- **Financing margins.** Long-term debt issuance falls by 1.07% and equity issuance falls by 0.90% of total assets. Short-term debt is essentially unchanged.
- **Cash use.** Cash holdings fall by 1.12% of total assets, consistent with firms replacing external finance with internal funds.
- **Payouts.** Dividends and share repurchases do not meaningfully change, suggesting the shock mainly affects investment and financing rather than shareholder payouts.
- **Heterogeneity.** Effects are strongest for smaller firms, firms with less analyst coverage, firms with larger increases in information asymmetry, and firms classified as financially constrained.
- **Robustness.** The results are not driven by underwriting relationships, time clustering of broker disappearances, matching methodology, firm life cycle, broker closures versus mergers, or low-quality analysts.
- **Economic takeaway.** Analysts are not just stock-market commentators. They are information intermediaries whose disappearance can reduce firms' access to capital and lead to real reductions in corporate investment.

2 Data and measurement

2.1 Why analyst coverage is a useful laboratory

- Analysts produce earnings forecasts, recommendations, price targets, and other information that helps investors evaluate firms.
- If analyst coverage falls, public information about the firm falls, information asymmetry rises, and external financing can become more costly.
- Broker closures and broker mergers are useful because they can force analysts to disappear for broker-level reasons rather than firm-level reasons.

Interpretation: The paper studies the second step of the information channel. Kelly and Ljungqvist show that broker closures increase information asymmetry and cost of capital. Derrien and Kecskes test whether that increase changes real corporate policies.

2.2 Treatment events: broker closures and broker mergers

- Broker disappearance dates are identified using I/B/E/S, press releases, and Securities Industry Association Yearbooks.
- A firm is treated if a broker that covered the firm disappears and the analyst covering the firm no longer issues earnings estimates after the disappearance.
- For broker closures, the paper keeps firms for which the analyst disappears from I/B/E/S during the year after the broker disappearance.
- For broker mergers, the paper keeps firms covered by both the target broker and the acquirer before the merger and for which one analyst disappears.
- The design excludes cases where coverage was already “stopped” before the broker disappearance, reducing the risk that analysts end coverage because they anticipate firm-specific changes.

Interpretation: The shock is a loss of analyst coverage induced by broker-level events, not a voluntary analyst decision to stop covering a firm.

2.3 Treatment and control sample

- The sample includes publicly traded U.S. operating firms that are not financials or utilities.
- Firms must have traded for at least one year before the broker disappearance and must have Compustat data in both year -1 and year $+1$.
- The final treatment sample has 1,724 firms that lose one analyst between 1994 and 2008.
- There are 1,724 matched control firms.
- There are 52 broker disappearances: 19 broker closures and 33 broker mergers.

Interpretation: Matching treatment and control firms makes the comparison less dependent on functional-form assumptions and reduces the concern that treatment firms differ systematically before the shock.

2.4 Matching variables

Candidate controls must have the same two-digit SIC code as the treatment firm and must be in the same quintile of:

- total assets;
- Tobin’s Q ;
- cash flow.

The selected control firm is then chosen to have the closest analyst coverage, with remaining ties broken by closeness in total assets, Q , and cash flow.

Interpretation: The matching variables are standard determinants of investment and financing. Matching on analyst coverage also ensures that losing one analyst has a comparable relative meaning for treatment and control firms.

2.5 Timing of before and after variables

- Because broker disappearance dates in news sources and I/B/E/S do not always match exactly, the paper measures variables before and after with a buffer.
- The “before” measurement is taken three months before the broker disappearance date.
- The “after” measurement is taken three months after the broker disappearance date.
- Year +1 is the year immediately after the broker disappearance date plus three months.

Interpretation: The buffer reduces ambiguity about whether the analyst loss already occurred when a variable is measured.

2.6 Corporate policy variables

All corporate policy variables are scaled by total assets. The main groups are:

- **Investment:** capital expenditures, research and development expenditures, acquisitions expenditures, and total investment.
- **Financing:** change in short-term debt, change in long-term debt, equity issuance, and total financing.
- **Payouts:** dividends, share repurchases, and total payouts.
- **Internal funds:** change in cash holdings.

Interpretation: The investment variables measure real activity; the financing variables measure external capital raising; the cash variable captures substitution toward internal liquidity.

2.7 Analysts’ expectations variables

The paper uses analyst expectations to check whether treatment and control firms differ before the shock. The expectations variables include:

- earnings estimates for the next fiscal year as a percent of stock price;
- investment recommendations on a five-point scale;
- long-term earnings growth rate estimates for the next five years;
- price targets for the next year as a percent of stock price.

Interpretation: If analysts expected treatment firms to reduce investment before broker disappearance, the analyst loss might not be exogenous. The paper finds expectations are broadly similar across treatment and control firms.

2.8 Information asymmetry and financial constraints variables

The paper studies whether effects are stronger when the loss of an analyst is more costly. It uses:

- information-asymmetry proxies: bid-ask spread, Amihud illiquidity, zero/missing return days, earnings-announcement surprise, and earnings-announcement volatility;
- financial-constraint proxies: a composite proxy from Almeida, Campello, and Weisbach, plus firm- and industry-level proxies from Rajan and Zingales.

Interpretation: These heterogeneity tests are mechanism tests. If the results are driven by information asymmetry and financing constraints, effects should be bigger exactly for these groups.

3 Models and empirical specifications

3.1 Mechanism in one line

Broker disappearance $\Rightarrow$ loss of analyst coverage $\Rightarrow$ higher information asymmetry $\Rightarrow$ higher cost of capital $\Rightarrow$ lower investment

Interpretation: The empirical tests focus on whether the last step is visible in corporate policies.

3.2 Main difference-in-differences estimand

For firm i and outcome Y , define the change from before to after as

$$\Delta Y_i = Y_{i,+1} - Y_{i,-1}.$$

The difference-in-differences estimate is

$$\hat{\beta}_{\text{DID}} = (\bar{Y}_{T,+1} - \bar{Y}_{T,-1}) - (\bar{Y}_{C,+1} - \bar{Y}_{C,-1}).$$

Interpretation: This compares the change for firms that lose an analyst with the change for otherwise similar firms that do not.

3.3 Equivalent pooled regression

For a before-after panel, the pooled specification is

$$Y_{i,t} = \alpha + \beta(\text{Treat}_i \times \text{After}_t) + \gamma \text{Treat}_i + \delta \text{After}_t + \varepsilon_{i,t}. \quad (1)$$

The coefficient of interest is β .

Interpretation: A negative β for investment or financing means that treated firms reduce that policy more than control firms after the loss of analyst coverage.

3.4 Event-time visual check

For each event year $s \in \{-3, -2, -1, +1, +2, +3\}$, the paper plots

$$\bar{Y}_{T,s} - \bar{Y}_{C,s}.$$

Interpretation: If treatment and control firms were already diverging before year -1 , the difference-in-differences interpretation would be weaker. The event-time figure shows that the main changes occur after the analyst loss.

3.5 Heterogeneity by market capitalization and analyst coverage

For quintile q , the paper estimates

$$Y_{i,t} = \alpha_q + \beta_q(\text{Treat}_i \times \text{After}_t) + \gamma_q \text{Treat}_i + \delta_q \text{After}_t + \varepsilon_{i,t}. \quad (2)$$

Interpretation: Smaller firms and firms with fewer analysts should be more affected because one analyst is more important for their information environment.

3.6 Triple-difference specification

For a group indicator G_i such as “big increase in information asymmetry” or “financially constrained,” the triple-difference logic is

$$\hat{\theta} = [(\Delta \bar{Y}_{T,G=1} - \Delta \bar{Y}_{C,G=1}) - (\Delta \bar{Y}_{T,G=0} - \Delta \bar{Y}_{C,G=0})]. \quad (3)$$

Interpretation: A negative θ for investment or financing means the analyst-loss effect is larger for the group predicted by the mechanism.

3.7 Underwriting robustness

The paper repeats the main difference-in-differences tests separately for:

- firms for which the disappearing broker is not an underwriter;
- firms for which the broker is a debt underwriter;
- firms for which the broker is an equity underwriter.

Interpretation: If results were caused by loss of an underwriting relationship rather than loss of an analyst, they should disappear among firms without underwriting ties. They do not.

4 Identification and empirical design

4.1 What identifies the main effect

- Treatment comes from broker closures and mergers that remove analysts from firms.
- The analysis compares treatment firms with closely matched control firms.
- The main identifying assumption is parallel trends: absent the analyst loss, treatment and control firms would have changed investment and financing similarly.

Interpretation: The design is a natural experiment rather than a randomized experiment. Its credibility depends on showing that treatment and control firms look similar before the shock and diverge only after the shock.

4.2 Why broker closures and mergers are useful

- Broker closures and mergers are broker-level events.
- Analysts are terminated or redundant because of the broker event, not necessarily because of the firm's own future policies.
- The paper removes cases where analyst coverage was stopped before the broker event.

Interpretation: This helps separate the loss of information production from firm-level investment opportunities.

4.3 Pre-treatment balance

- Table I shows that treatment and control firms are similar in matching variables, corporate policy variables, and most analyst expectation variables.
- The few significant analyst-expectation differences are small and point in opposite directions.
- Figure 2 shows analyst coverage for treatment firms falls sharply around the broker disappearance, while control firms do not experience the same loss.

Interpretation: The balance tests support the exogeneity of the analyst-loss shock.

4.4 Parallel trends

- Figure 3 plots event-time differences between treatment and control firms.
- Investment and financing are fairly flat before year -1 and fall mainly between year -1 and year $+1$.
- Payouts barely move.
- Cash holdings fall after the analyst loss.

Interpretation: The visual evidence supports the paper’s parallel-trends assumption.

4.5 Mechanism checks

- Effects are stronger for small firms and firms with few analysts.
- Effects are stronger when information asymmetry increases more.
- Effects are stronger for financially constrained firms.
- Firms reduce equity and long-term debt more than short-term debt, consistent with switching away from information-sensitive financing.

Interpretation: The heterogeneity aligns with the cost-of-capital mechanism rather than a generic firm shock.

4.6 What the paper cannot fully prove

- Broker disappearances are not randomly assigned.
- Broker disappearances can be clustered in calendar time.
- Treatment firms may differ from controls in unobserved ways not captured by matching.
- The paper measures a loss of analyst coverage, not the exact change in a firm’s cost of capital.
- The estimated effects are local to firms that lose one analyst due to broker disappearance.

Interpretation: The paper’s strength is the convergence of evidence: main DID, event-time plots, mechanism heterogeneity, and robustness tests all point in the same direction.

5 Tables and figures

This section preserves the paper’s main visual evidence. Adjacent figures and tables are grouped where possible to keep the notes compact.

5.1 Figure 1 + Figure 2: treatment timing and analyst coverage loss

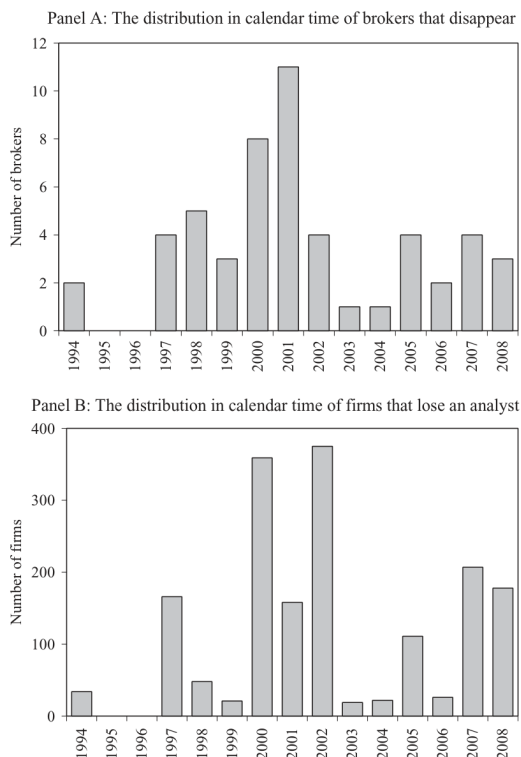
Read: Figure 1:

- Broker disappearances are spread across 1994–2008, with some clustering around 2000–2002.
- Firms that lose an analyst are more strongly clustered than brokers, because some broker disappearances affect many covered firms.
- This motivates robustness tests that address time clustering and large broker events.

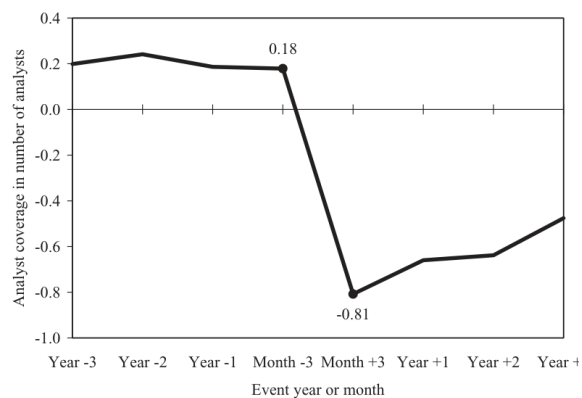
Read: Figure 2:

- Treatment firms and control firms have similar analyst coverage before the event.
- Treatment firms lose about 0.81 analysts by three months after the broker disappearance.
- Control firms do not experience a comparable drop.

Economic message: The design has a clear first stage: broker disappearance causes a sharp analyst coverage shock for treatment firms.



(a) Figure 1: broker disappearances and firms losing analysts.



(b) Figure 2: analyst coverage in event time.

5.2 Table I: descriptive statistics

Read:

- Treatment and control firms are closely matched on total assets, Q , cash flow, and number of analysts.
- Corporate policies are also similar before the event: investment, financing, payouts, and cash changes are not meaningfully different.
- Earnings estimates and recommendations are not significantly different.
- Long-term earnings growth and price targets differ statistically, but in opposite directions and economically small ways.

Economic message: The matching procedure produces a credible control group. The main post-event changes are unlikely to be driven by large pre-event differences.

Table I
Descriptive Statistics

This table presents descriptive statistics that compare treatment firms and control firms. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q, cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. All corporate policy variables are scaled by total assets. Analysts' expectations variables are computed as the mean expectations of all analysts covering the firm, and they comprise the following: earnings estimates for the next fiscal year measured as a percent of the stock price, investment recommendations measured on a five-point scale (a higher value of which means a more favorable recommendation), long-term earnings growth rate estimates for the next 5 years, and price targets for the next year measured as a percent of the stock price. All variables are measured during the year before the broker disappearance date.

	25 th Percentile		Median		75 th Percentile		<i>p</i> -value of Test of Equality of Medians	<i>p</i> -value of Test of Equality of Distributions
	Treatment Firms	Control Firms	Treatment Firms	Control Firms	Treatment Firms	Control Firms		
Matching variables								
Total assets (\$M)	588	586	2,329	2,246	9,016	7,867	0.658	0.554
Q	1.28	1.32	1.81	1.88	2.94	3.06	0.143	0.217
Cash flow	4.82%	5.36%	9.28%	9.74%	14.17%	14.86%	0.143	0.118
Number of analysts	11.0	10.0	18.0	18.0	24.0	25.0	0.918	0.583
Investment variables								
Capital expenditures	2.60%	2.47%	4.82%	4.65%	8.50%	8.23%	0.261	0.372
Research and development exp.'s	0.00%	0.00%	0.23%	0.18%	5.78%	5.29%	0.812	0.699
Acquisitions expenditures	0.00%	0.00%	0.00%	0.04%	2.14%	2.22%	0.261	0.670
Total investment	6.13%	6.16%	10.52%	10.18%	17.33%	16.88%	0.433	0.670
Financing variables								
Change in short-term debt	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.927	0.974
Change in long-term debt	− 1.05%	− 0.98%	0.00%	0.00%	3.21%	3.25%	0.781	0.641
Equity issuance	0.19%	0.22%	0.77%	0.81%	2.28%	2.29%	0.433	0.350
Total financing	− 0.20%	− 0.15%	2.14%	2.10%	6.54%	7.15%	0.759	0.860

(Continued)

Table I, part 1: descriptive statistics.

Table I—Continued

	25 th Percentile		Median		75 th Percentile		<i>p</i> -value of Test of Equality of Medians	<i>p</i> -value of Test of Equality of Distributions
	Treatment Firms	Control Firms	Treatment Firms	Control Firms	Treatment Firms	Control Firms		
Payout variables								
Dividends	0.00%	0.00%	0.00%	0.00%	1.35%	1.36%	0.340	0.810
Share repurchases	0.00%	0.00%	0.14%	0.08%	3.14%	3.17%	0.233	0.670
Total payouts	0.00%	0.00%	1.23%	1.15%	5.28%	5.05%	0.586	0.836
Change in cash holdings	− 1.10%	− 0.84%	0.15%	0.26%	2.35%	2.66%	0.610	0.117
Analysts' expectations variables								
Earnings estimates	2.6%	2.7%	4.9%	4.8%	7.1%	6.8%	0.584	0.203
Investment recommendations	3.5	3.5	3.8	3.9	4.2	4.2	0.185	0.181
Long-term earnings growth rate est.'s	11.4%	12.1%	15.3%	16.1%	22.9%	23.0%	0.028	0.015
Price targets	12.3%	11.1%	29.6%	24.2%	60.6%	50.3%	0.001	0.000

Table I, part 2: payout variables and analyst expectations.

5.3 Table II + Figure 3: main real effects and event-time behavior

Read: Table II:

- Total investment falls by 1.92% of total assets.
- Capital expenditures fall by 0.67%, R&D by 0.21%, and acquisitions by 0.97%.
- Total financing falls by 2.02% of total assets.
- Long-term debt falls by 1.07% and equity issuance by 0.90%; short-term debt is unchanged.

- Cash holdings fall by 1.12%; payouts are not significantly affected.

Read: Figure 3:

- Investment and financing differences are fairly stable before the event.
- The decline happens mainly from year -1 to year $+1$.
- The cash-holdings decline also occurs after the analyst loss.

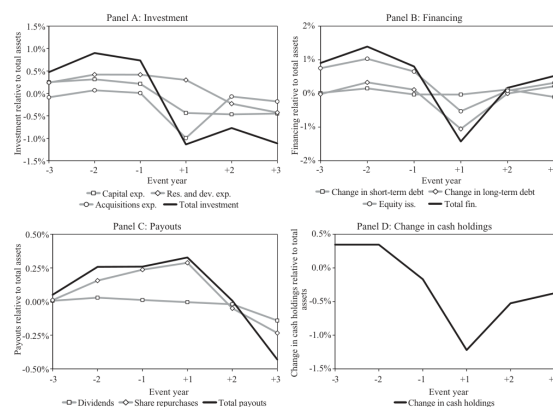
Economic message: The results show real corporate responses to an information shock: firms cut investment and external financing and draw down cash.

Table II
The Effect of a Decrease in Analyst Coverage on Corporate Policies

This table presents the change in corporate policies caused by the loss of an analyst. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. For each corporate policy variable, the mean change from the year before the decrease in analyst coverage to the year after is computed for treatment firms (the treatment difference), control firms (the control difference), and the difference between treatment firms and control firms (the difference-in-differences). All corporate policy variables are scaled by total assets. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. Statistical significance is only tabulated for the mean of the difference-in-differences.

	Mean Treatment Difference (Year +1 vs. Year -1)	Mean Control Difference (Year +1 vs. Year -1)	Mean of Diff-in-Diffs (Treatments vs. Controls)	<i>t</i> -statistic for Difference-in- Differences
Investment				
Capital expenditures	-0.01%	0.62%	-0.67%***	-4.73
Research and dev. exp.'s	0.40%	0.60%	-0.21%**	-2.27
Acquisitions expenditures	0.09%	1.11%	-0.97%***	-3.36
Total investment	0.41%	2.41%	-1.92%***	-5.18
Financing				
Change in short-term debt	0.06%	0.06%	0.00%	0.02
Change in long-term debt	-0.17%	0.97%	-1.07%***	-2.79
Equity issuance	-1.13%	-0.10%	-0.90%***	-2.88
Total financing	-1.34%	0.89%	-2.02%***	-4.00
Payouts				
Dividends	0.10%	0.12%	-0.02%	-1.12
Share repurchases	0.15%	0.08%	0.08%	0.48
Total payouts	0.30%	0.23%	0.11%	0.61
Change in cash holdings	-0.06%	1.03%	-1.12%***	-2.62

(a) Table II: main difference-in-differences effects.



(b) Figure 3: corporate policies in event time.

5.4 Table III: effects by market capitalization and analyst coverage

Read:

- In the smallest market-cap quintile, total investment falls by 2.44% and total financing by 5.69%.
- For the largest firms, the corresponding effects are much smaller and statistically insignificant.
- For firms with the least analyst coverage, total investment falls by 2.69% and total financing by 4.76%.
- The effect weakens as initial analyst coverage increases.

Economic message: One analyst matters more when the firm is small or thinly covered. This supports the information-asymmetry channel.

Table III
The Effect of a Decrease in Analyst Coverage on Corporate Policies
Conditional upon Market Capitalization and Analyst Coverage

This table presents the mean change in corporate policies caused by the loss of an analyst conditional upon market capitalization and analyst coverage. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. For each corporate policy variable, one pooled regression is run using treatment firms and control firms before and after the decrease in analyst coverage. There is one “before” observation and one “after” observation for each treatment firm and each control firm. Firms are sorted into quintiles based on the value of conditioning variables for treatment firms and control firms before the decrease in analyst coverage. For each quintile, there is a constant term, a “treatment firm” dummy variable, an “after” dummy variable, and an interaction between the “treatment firm” dummy variable and the “after” dummy variable. All corporate policy variables are scaled by total assets. All conditioning variables are measured during the year before the decrease in analyst coverage. Only the interaction terms (the differences-in-differences) are tabulated. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Panel A: Mean Differences-in-Differences Conditional upon Market Capitalization					
	Q1 (smallest)	Q2	Q3	Q4	Q5 (biggest)
Investment					
Capital expenditures	−0.83	−1.01	−0.74	−0.53	0.05
Res. and dev. expenditures	0.07	−0.22	−0.21	0.12	−0.27
Acquisitions expenditures	−1.25*	−0.96	−1.17*	−1.03	−0.62
Total investment	−2.44**	−2.34**	−2.03*	−1.47	−0.79
Financing					
Change in short-term debt	0.31	−0.50**	0.20	0.08	−0.03
Change in long-term debt	−2.09**	−1.91**	−1.98**	−0.23	0.46
Equity issuance	−4.28***	−0.16	−0.75	−1.10	0.51
Total financing	−5.69***	−2.62**	−2.58**	−1.16	1.16
Payouts					
Dividends	−0.08	0.03	−0.04	0.08	−0.05
Share repurchases	−0.03	0.05	0.16	0.09	0.03
Total payouts	−0.05	0.19	0.08	0.16	0.00
Change in cash holdings	−1.91**	−0.81	−1.19	−1.35	−0.11
Panel B: Mean Differences-in-Differences Conditional upon Analyst Coverage					
	Q1 (least)	Q2	Q3	Q4	Q5 (most)
Investment					
Capital expenditures	−0.75	−1.05	−0.58	−0.49	−0.43
Res. and dev. expenditures	0.11	−0.38	−0.05	−0.33	0.00
Acquisitions expenditures	−1.61***	−1.58**	−0.83	−0.29	−0.74
Total investment	−2.69**	−3.00**	−1.47	−1.04	−1.24
Financing					
Change in short-term debt	0.01	0.52**	−0.15	−0.42	−0.04
Change in long-term debt	−0.83	−2.38***	−0.14	−1.00	−1.47*
Equity issuance	−4.07***	−0.05	−1.49*	−0.73	0.74
Total financing	−4.76***	−1.70	−1.66	−1.94	−0.83

Table III, part 1: market capitalization and analyst coverage heterogeneity.

Table III—Continued

Panel B: Mean Differences-in-Differences Conditional Upon Analyst Coverage					
	Q1 (least)	Q2	Q3	Q4	Q5 (most)
Payouts					
Dividends	−0.04	−0.03	−0.02	−0.00	0.01
Share repurchases	0.29	−0.19	0.33	−0.27	0.06
Total payouts	0.34	−0.21	0.27	−0.34	0.23
Change in cash holdings	−1.76**	−0.90	−1.45	0.53	−1.49

Table III, part 2: continuation of analyst coverage panel.

5.5 Table IV: effects by increase in information asymmetry

Read:

- Firms with a big increase in bid-ask spreads reduce total investment 3.16 percentage points more than

firms with a small increase.

- The same comparison shows total financing falling 7.31 percentage points more.
- Using Amihud illiquidity, the corresponding extra declines are 4.86 percentage points for investment and 7.70 percentage points for financing.
- Cash holdings fall more for firms with larger information-asymmetry increases.

Economic message: This is a direct mechanism test. When the analyst loss produces a bigger information problem, the real response is larger.

Table IV
The Effect of a Decrease in Analyst Coverage on Corporate Policies Conditional upon the Increase in Information Asymmetry

This table presents the change in corporate policies caused by the loss of an analyst conditional upon the increase in information asymmetry. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. For each corporate policy variable, a mean difference-in-differences-in-differences is computed: the difference between the year after the decrease in analyst coverage and the year before, the difference between treatment firms and control firms, and the difference between firms that have a big change in information asymmetry and firms that have a small change in information asymmetry. All corporate policy variables are scaled by total assets. Firms in the top tercile of the change in information asymmetry are classified as having a big change and firms in the bottom tercile are classified as having a small change. The construction of the information asymmetry variables is detailed in Panel B of Appendix Table AI. All conditioning variables are measured as differences-in-differences: the difference between the year after versus the year before and the difference between treatment firms versus control firms. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Differences-in-Differences-in-Differences		Big Change in Bid-Ask Spread ($N = 521$) vs. Small ($N = 522$)	Big Change in Amihud Illiquidity Measure ($N = 567$) vs. Small ($N = 568$)	Big Change in Returns Ratio ($N = 571$) vs. Small ($N = 573$)	Big Change in Earnings Announcement Surprise ($N = 539$) vs. Small ($N = 540$)	Big Change in Earnings Announcement Volatility ($N = 556$) vs. Small ($N = 557$)
Investment						
Capital expenditures	Mean:	- 1.11%***	- 1.69%***	- 0.92%**	- 0.24%	- 0.67%*
	t -stat:	(- 2.76)	(- 4.50)	(- 2.47)	(- 0.63)	(- 1.82)
Research and dev. exp.'s	Mean:	- 0.44%*	- 0.47%*	- 0.14%	- 0.26%	- 0.12%
	t -stat:	(- 1.74)	(- 1.93)	(- 0.57)	(- 1.08)	(- 0.49)
Acquisitions expenditures	Mean:	- 1.61%**	- 2.13%***	- 1.54%**	- 1.83%**	- 0.48%
	t -stat:	(- 2.18)	(- 2.87)	(- 2.05)	(- 2.52)	(- 0.65)
Total investment	Mean:	- 3.16%***	- 4.86%***	- 2.91%***	- 2.59%***	- 1.32%
	t -stat:	(- 3.19)	(- 4.97)	(- 2.99)	(- 2.70)	(- 1.38)

(Continued)

Table IV, part 1: investment effects by increase in information asymmetry.

Table IV—Continued

Differences-in-Differences-in-Differences		Big Change in Bid-Ask Spread ($N = 521$) vs. Small ($N = 522$)	Big Change in Amihud Illiquidity Measure ($N = 567$) vs. Small ($N = 568$)	Big Change in Returns Ratio ($N = 571$) vs. Small ($N = 573$)	Big Change in Earnings Announcement Surprise ($N = 539$) vs. Small ($N = 540$)	Big Change in Earnings Announcement Volatility ($N = 556$) vs. Small ($N = 557$)
Financing						
Change in short-term debt	Mean:	− 0.07%	− 0.26%	− 0.50%	0.03%	0.07%
	t -stat:	(− 0.22)	(− 0.89)	(− 1.48)	(0.11)	(0.22)
Change in long-term debt	Mean:	− 2.36%**	− 2.14%**	− 1.50%	− 2.20%**	− 1.47%
	t -stat:	(− 2.34)	(− 2.10)	(− 1.55)	(− 2.21)	(− 1.49)
Equity issuance	Mean:	− 5.06%***	− 5.09%***	− 3.79%***	− 2.51%***	− 0.33%
	t -stat:	(− 5.60)	(− 5.92)	(− 4.78)	(− 3.06)	(− 0.38)
Total financing	Mean:	− 7.31%***	− 7.70%***	− 5.60%***	− 5.00%***	− 1.01%
	t -stat:	(− 5.36)	(− 5.70)	(− 4.41)	(− 3.84)	(− 0.76)
Payouts						
Dividends	Mean:	− 0.10%**	− 0.13%***	− 0.07%	− 0.11%**	− 0.02%
	t -stat:	(− 2.08)	(− 2.85)	(− 1.57)	(− 2.44)	(− 0.47)
Share repurchases	Mean:	− 0.34%	− 0.07%	0.01%	− 0.39%	0.27%
	t -stat:	(− 0.87)	(− 0.19)	(0.02)	(− 1.02)	(0.65)
Total payouts	Mean:	− 0.34%	− 0.07%	− 0.12%	− 0.50%	0.17%
	t -stat:	(− 0.83)	(− 0.18)	(− 0.28)	(− 1.22)	(0.37)
Change in cash holdings	Mean:	− 3.47%***	− 3.08%***	− 3.39%***	− 1.83%	− 2.03%*
	t -stat:	(− 2.93)	(− 2.68)	(− 3.07)	(− 1.64)	(− 1.74)

Table IV, part 2: financing, payouts, and cash effects.

5.6 Table V: effects by financial constraints

Read:

- Using the Rajan-Zingales firm-level constraint proxy, constrained firms reduce total investment 4.98 percentage points more than unconstrained firms.
- Using the same proxy, constrained firms reduce total financing 4.39 percentage points more.
- Equity issuance and long-term debt are particularly sensitive for constrained firms.
- The Almeida-Campello-Weisbach and industry-level Rajan-Zingales proxies give directionally similar evidence, though not every coefficient is significant.

Economic message: The analyst-coverage shock matters most where external finance is already difficult. That is exactly what a cost-of-capital channel predicts.

Table V
The Effect of a Decrease in Analyst Coverage on Corporate Policies
Conditional upon Financial Constraints

This table presents the change in corporate policies caused by the loss of an analyst conditional upon financial constraints. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. For each corporate policy variable, a mean difference-in-differences-in-differences is computed: the difference between the year after the decrease in analyst coverage and the year before, the difference between treatment firms and control firms, and the difference between firms that are financial constrained and firms that are not financially constrained. All corporate policy variables are scaled by total assets. The construction of the financial constraints variables is detailed in Panel C of Appendix Table AI. All conditioning variables are measured using only treatment firms, and they are measured during the year before the decrease in analyst coverage. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively

Differences-in-Differences-in-Differences	Constrained ($N = 358$) vs. Unconstrained ($N = 435$) Based on the Composite Proxy from Almeida, Campello, and Weisbach (2004)		Constrained ($N = 862$) vs. Unconstrained ($N = 862$) Based on the Firm-Level Proxy from Rajan and Zingales (1998)		Constrained ($N = 1,034$) vs. Unconstrained ($N = 294$) Based on the Industry-Level Proxy from Rajan and Zingales (1998)	
	Mean	t -statistic	Mean	t -statistic	Mean	t -statistic
Investment						
Capital expenditures	-0.39%	-0.89	-1.46%***	-5.17	-1.23%***	-4.26
Research and dev. exp.'s	0.09%	0.32	-0.16%	-0.87	-0.26%	-1.36
Acquisitions expenditures	-0.41%	-0.51	-3.19%***	-5.59	-0.26%	-0.40
Total investment	-0.81%	-0.74	-4.98%***	-6.79	-1.86%**	-2.36

(Continued)

Table V, part 1: investment effects by financial constraints.

Table V—Continued

Differences-in-Differences-in-Differences	Constrained ($N = 358$) vs. Unconstrained ($N = 435$) Based on the Composite Proxy from Almeida, Campello, and Weisbach (2004)		Constrained ($N = 862$) vs. Unconstrained ($N = 862$) Based on the Firm-Level Proxy from Rajan and Zingales (1998)		Constrained ($N = 1,034$) vs. Unconstrained ($N = 294$) Based on the Industry-Level Proxy from Rajan and Zingales (1998)	
	Mean	t -statistic	Mean	t -statistic	Mean	t -statistic
Financing						
Change in short-term debt	0.33%	0.79	-0.57%**	-2.19	0.11%	0.33
Change in long-term debt	-1.49%	-1.29	-2.64%***	-3.46	-1.21%	-1.41
Equity issuance	-2.71%***	-2.73	-1.06%*	-1.69	-0.64%	-1.01
Total financing	-3.65%**	-2.43	-4.39%***	-4.36	-1.89%*	-1.76
Payouts						
Dividends	-0.05%	-1.03	-0.06%	-1.56	0.07%	1.50
Share repurchases	0.34%	0.76	-0.07%	-0.22	0.43%	1.09
Total payouts	0.18%	0.40	-0.09%	-0.27	0.56%	1.33
Change in cash holdings	-2.51%*	-1.89	-0.49%	-0.57	-1.25%	-1.40

Table V, part 2: financing and cash effects by financial constraints.

5.7 Table VI + Table VII: analyst quality and underwriting robustness

Read: Table VI:

- The brokers and analysts that disappear are not low quality.
- Relative earnings-estimate accuracy is slightly above average: broker mean accuracy is 0.520 and analyst mean accuracy is 0.536.

- Analysts that disappear have expectations broadly similar to other analysts covering the same treatment firms.

Read: Table VII:

- Most firms have no underwriting relationship with the disappearing broker.
- Among firms where the broker is not an underwriter, total investment falls by 1.71% and total financing by 1.87%.
- The result therefore cannot be explained by loss of an underwriting relationship alone.

Economic message: The paper rules out two important alternative stories: that the disappearing analysts were unimportant low-quality analysts, or that the effect is really about losing an underwriter.

Table VI
Broker and Analyst Quality

This table presents results on the quality of brokers and analysts. Panel A presents the relative earnings estimate accuracy of brokers and analysts that disappear. Panel B presents analysts' expectations for analysts that disappear compared to other analysts. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. Relative earnings estimate accuracy is constructed by first computing the accuracy rank—on a scale of zero to one—across all brokers that cover a firm and then computing the mean accuracy rank—also on a scale of zero to one—of a broker across all firms that the broker covers. Analysts' expectations comprise the following: earnings estimates for the next fiscal year measured as a percent of the stock price, investment recommendations measured on a five-point scale (a higher value of which means more a favorable recommendation), long-term earnings growth rate estimates for the next 5 years, and price targets for the next year measured as a percent of the stock price. Expectations of other analysts are computed as the mean of their expectations. They are only computed for other analysts if they are available for the analyst that disappears. All variables are measured during the year before the broker disappearance date.

Panel A: Relative Earnings Estimate Accuracy of Brokers and Analysts that Disappear				
	Number of Observations	Mean	Median	Mean Weighted by Number of Firms Covered
Brokers	71	0.520	0.539	0.535
Analysts	373	0.536	0.546	0.554

Panel B: Analysts' Expectations for Treatment Firms for Analysts that Disappear Compared to Other Analysts				
	Mean			
	Number of Observations	Analysts that Disappear	Other Analysts	p -value of Test of Equality
Earnings estimates	1,635	3.4%	3.8%	0.226
Investment recommendations	966	3.7	3.8	0.003
Long-term earn.'s gr. rate est.'s	698	18.4%	19.5%	0.043
Price targets	804	56.7%	60.3%	0.563

	Median			
	Number of Observations	Analysts that Disappear	Other Analysts	p -value of Test of Equality
Earnings estimates	1,635	4.7%	4.9%	0.235
Investment recommendations	966	4.0	3.8	0.000
Long-term earn.'s gr. rate est.'s	698	15.0%	16.5%	0.022
Price targets	804	26.2%	28.3%	0.441

(a) Table VI: broker and analyst quality.

Table VII
Underwriting Relationships

This table presents the change in corporate policies caused by the loss of an analyst conditional upon underwriting relationships. The sample comprises 1,724 treatment firms that lose an analyst between 1994 and 2008 because of broker closures and broker mergers, and the same number of control firms matched by industry, total assets, Q , cash flow, and analyst coverage. Both groups of firms are publicly traded U.S. operating firms, are not financials or utilities, and have been traded for at least 1 year. For each corporate policy variable, the mean difference-in-differences is computed conditional upon the broker not being an underwriter for the firm, conditional upon the broker being a debt underwriter for the firm, and conditional upon the broker being an equity underwriter for the firm. All corporate policy variables are scaled by total assets. A broker is an underwriter for the firm if it underwrites an offering during the 3 years before the broker disappearance date. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Differences-in-Differences	Broker Is Not Underwriter ($N = 1,537$)		Broker Is Debt Underwriter ($N = 136$)		Broker Is Equity Underwriter ($N = 58$)	
	Mean	t -statistic	Mean	t -statistic	Mean	t -statistic
Investment						
Capital expenditures	-0.64%***	-4.35	-1.20%**	-2.12	-1.16%	-1.00
Research and development expenditures	-0.17%*	-1.75	-0.56%**	-2.16	-0.39%	-0.42
Acquisitions expenditures	-0.85%***	-2.82	-2.37%***	-3.88	-1.35%	-0.60
Total investment	-1.71%***	-4.41	-4.20%***	-3.79	-3.92%	-1.21
Financing						
Change in short-term debt	0.00%	0.01	-0.16%	-0.29	0.25%	0.49
Change in long-term debt	-0.81%**	-2.01	-2.77%**	-2.03	-1.98%	-0.58
Equity issuance	-0.95%***	-2.93	0.54%	0.78	-4.76%*	-1.92
Total financing	-1.87%***	-3.54	-2.26%	-1.39	-5.50%	-1.30
Payouts						
Dividends	-0.01%	-0.72	-0.11%	-1.33	0.01%	0.37
Share repurchases	0.09%	0.52	-0.09%	-0.15	0.29%	0.35
Total payouts	0.14%	0.77	-0.35%	-0.54	0.35%	0.40
Change in cash holdings	-1.08%**	-2.42	1.60%*	1.76	-8.31%**	-2.08

(b) Table VII: underwriting relationships.

5.8 Appendix Table A.1: variable definitions

Read:

- The appendix defines investment, financing, payout, cash, information-asymmetry, and financial-constraint variables.
- The main variables are standard Compustat measures scaled by total assets.
- The information-asymmetry variables are based on liquidity, zero returns, and earnings-announcement reactions.

Economic message: The appendix clarifies that the main results are built from transparent accounting and market microstructure variables.

Variable	Definition in terms of composite variables (scaled by total assets)
Investment	
Capital expenditures	CAPX
Research and development expenditures	XRD
Acquisitions expenditures	AQC
Total investment	CAPX + XRD + AQC
Financing	
Change in short-term debt	DLCCH
Change in long-term debt	DLTIS-DLTR
Equity issuance	SSTK
Total financing	DLCCH + DLTIS-DLTR + SSTK
Payouts	
Dividends	DV
Share repurchases	PRSTKC
Total payouts	DV + PRSTKC
Change in cash holdings	CHECH
Panel B: Information Asymmetry Variables	
– Bid-ask spread: Mean during the year of the daily ask price minus the bid price all divided by the mean of the ask price and the bid price – Amihud illiquidity measure: Mean during the year of the daily absolute value of the stock return divided by the dollar value of trading volume – Ratio of zero and missing returns days to total days: Number of trading days with zero or missing returns during the year divided by the number of trading days during the year – Earnings announcement surprise: Mean during the year of the quarterly absolute value of the difference between actual earnings and expected earnings divided by the stock price – Earnings announcement volatility: Mean during the year of the quarterly volatility of the three-day market reaction to earnings announcements	

Appendix Table A.1: corporate policy and information-asymmetry variable definitions.

6 Short summary

- The paper asks whether analyst coverage has real effects on firm investment and financing.
- The causal variation comes from broker closures and mergers that cause analysts to disappear.
- The sample contains 52 broker disappearances, 1,724 treatment firms, and 1,724 matched control firms.
- Treatment and control firms are matched by industry, assets, Q , cash flow, and analyst coverage.
- Firms that lose an analyst reduce total investment by 1.92% of total assets.
- Firms that lose an analyst reduce total financing by 2.02% of total assets.
- Investment falls through capital expenditures, R&D, and acquisitions.
- Financing falls mainly through long-term debt and equity issuance, not short-term debt.
- Cash holdings fall, suggesting firms use internal funds after external financing becomes more costly.
- Payouts do not materially change.
- Event-time plots support parallel trends.
- Effects are larger for small firms and firms with fewer analysts.
- Effects are larger when information asymmetry rises more.
- Effects are larger for financially constrained firms.
- Analysts and brokers that disappear are not low-quality information providers.
- The results are not driven by underwriting relationships or broker-specific clustering.
- The paper’s broad message is that information intermediaries affect real corporate policies, not only securities prices.

7 Conclusion

7.1 Core conclusion

Derrien and Kecskes show that a decrease in analyst coverage causes firms to reduce investment and financing. The estimated effects are economically large: roughly two percent of total assets for both investment and financing. The results support the view that analysts reduce information asymmetry and help firms access external capital.

7.2 Economic intuition

Analysts produce information that makes outside investors more willing to finance firms. When a firm loses an analyst, outside investors know less about the firm's prospects. External capital becomes more expensive. The firm responds by cutting investment, reducing long-term debt and equity issuance, and drawing down cash. This pattern is strongest exactly where information problems and financing constraints are most severe.

7.3 Takeaway

The paper provides clean evidence that financial-market information affects real outcomes. Analysts are part of the financing technology of firms. Losing analyst coverage is therefore a financial shock with real consequences for investment, financing, and cash policy.